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The Zilog Z80



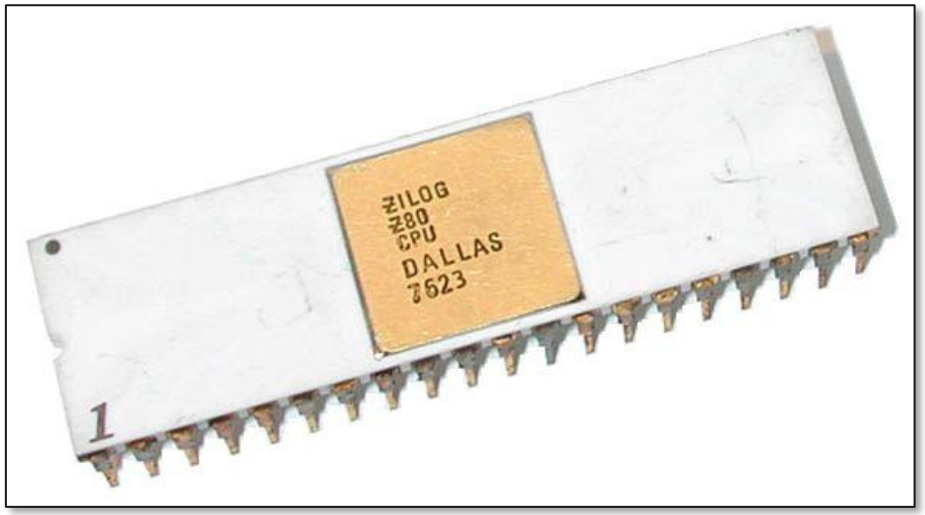
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Zilog Z80

Z80 Family of 8-bit Processor and Peripherals
1976 Zilog releases Z80 (compatible with 8080A)

- A Very Famous 8-bit Microprocessor (MPU)
- 4 μm depletion-mode NMOS → 2 μm
- 40-pin PDIP/CDIP
- Von Neumann Architecture
  - 16-bit address (64 kB Unified Memory)
  - 8-bit I/O address (256 I/O Devices)
- Single +5 V_{CC} Supply and 2.5 MHz → 20 MHz
- Interrupt support with banked registers
- 158 Documented Instructions

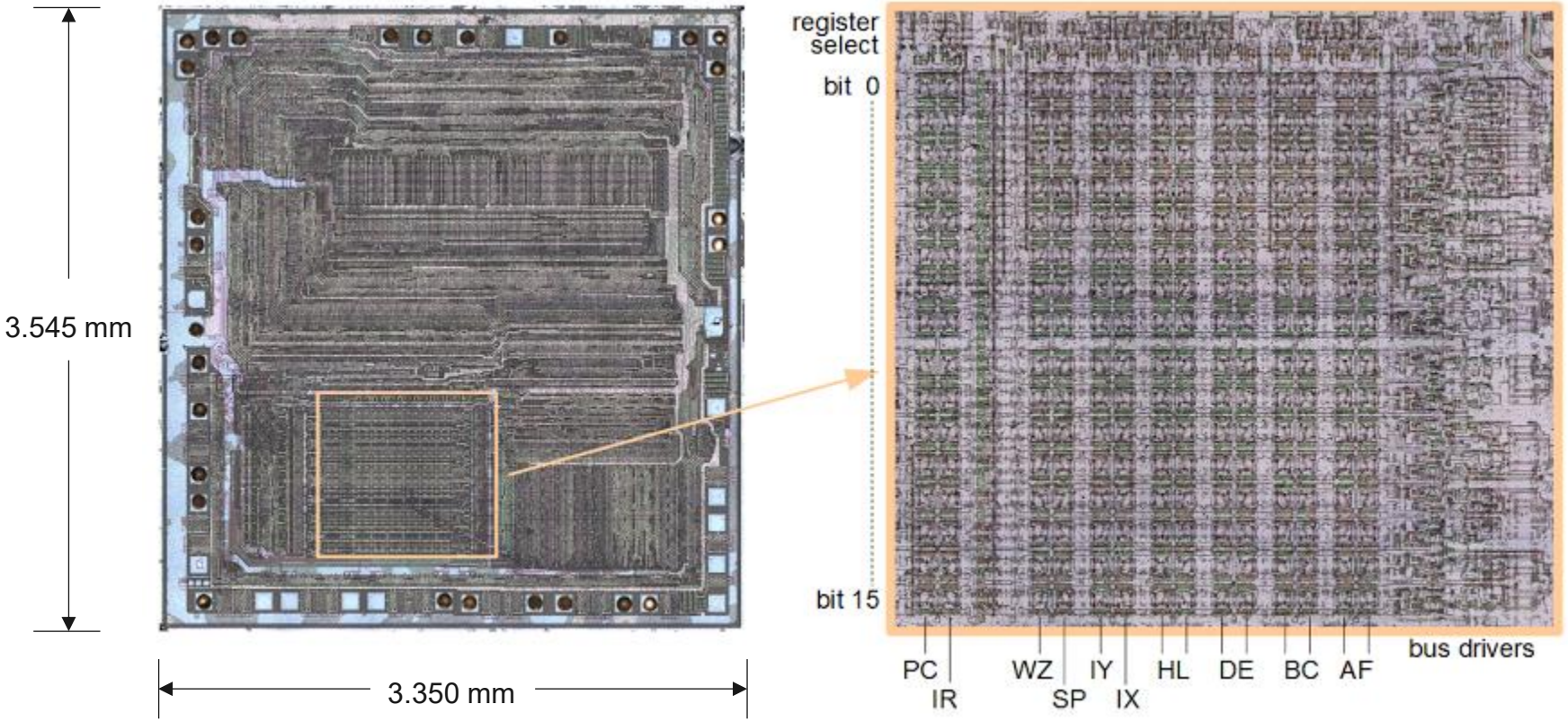
Produced until **2024** (48 years) !



By Gennadiy Shvets - Transferred from en.wikipedia to Commons., CC BY 2.5, https://commons.wikimedia.org/w/index.php?curid=1830432

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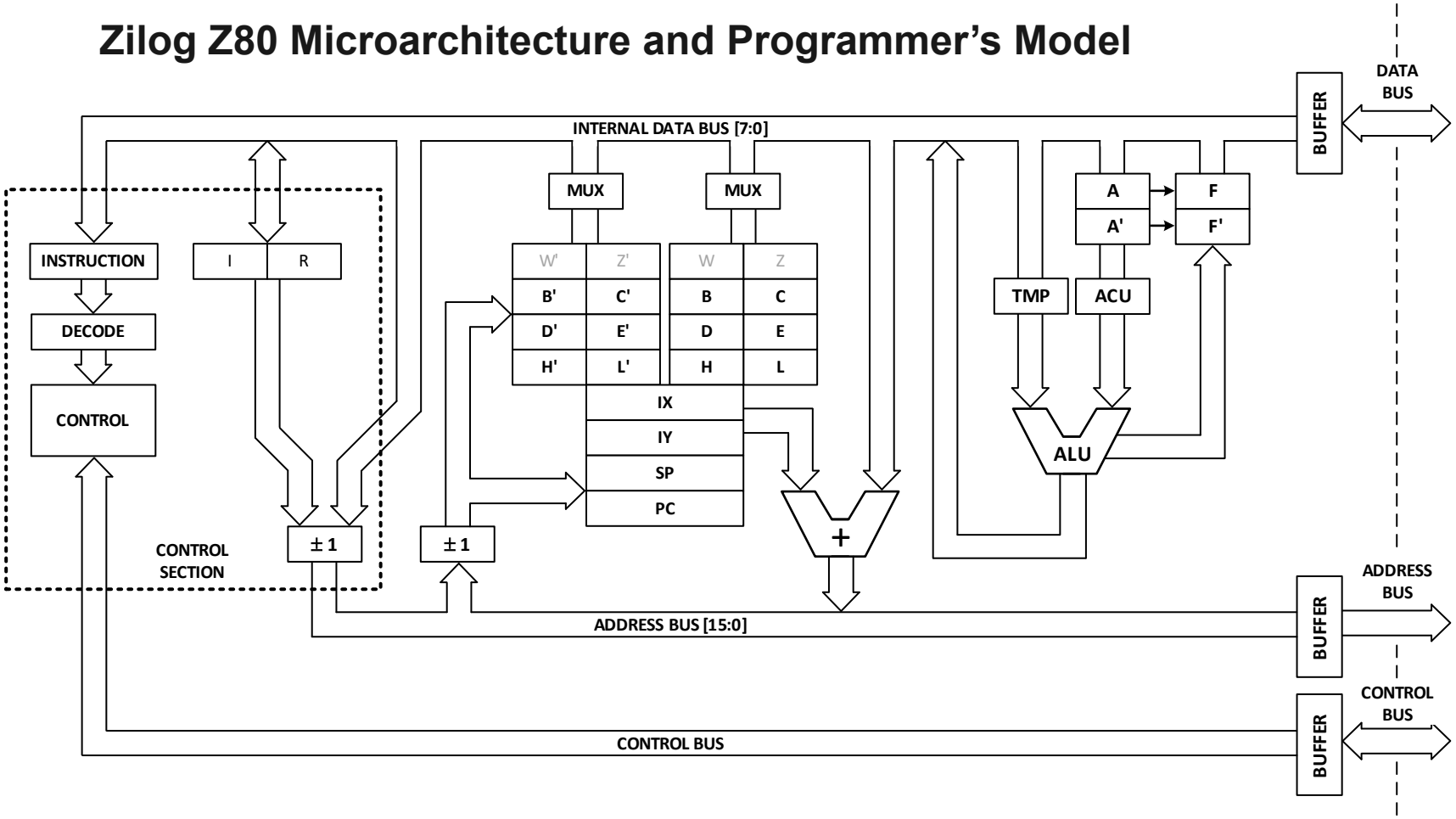
Zilog Z80 and Register Implementation



By Ken Shirriff, https://www.rightto.com

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Zilog Z80 Microarchitecture and Programmer's Model



Zilog Z80 registers

^{15} ^{14} ^{13} ^{12} ^{11} ^{10} ^{9} ^{8} ^{7} ^{6} ^{5} ^{4} ^{3} ^{2} ^{1} ^{0} *(bit position)*

Main registers

Accumulator (A)	Flags (F)	AF
B	C	BC
D	E	DE
H	L	HL

Alternate (shadow) registers

Accumulator' (A')	Flags' (F')	AF'
B'	C'	BC'
D'	E'	DE'
H'	L'	HL'

Index registers

Index X	IX
Index Y	IY
Stack Pointer	SP

Other registers

Interrupt vector	Refresh counter	I/R

Program counter

Program Counter	PC

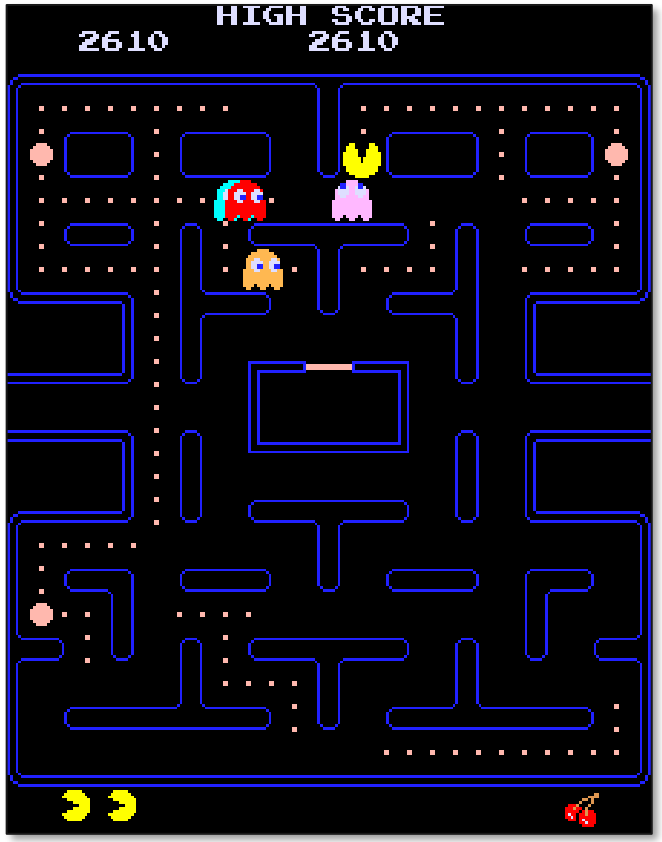
Status

S	Z	-	H	-	P/V	N	C	Flags
				IM	IFF1	IFF2		Interrupt

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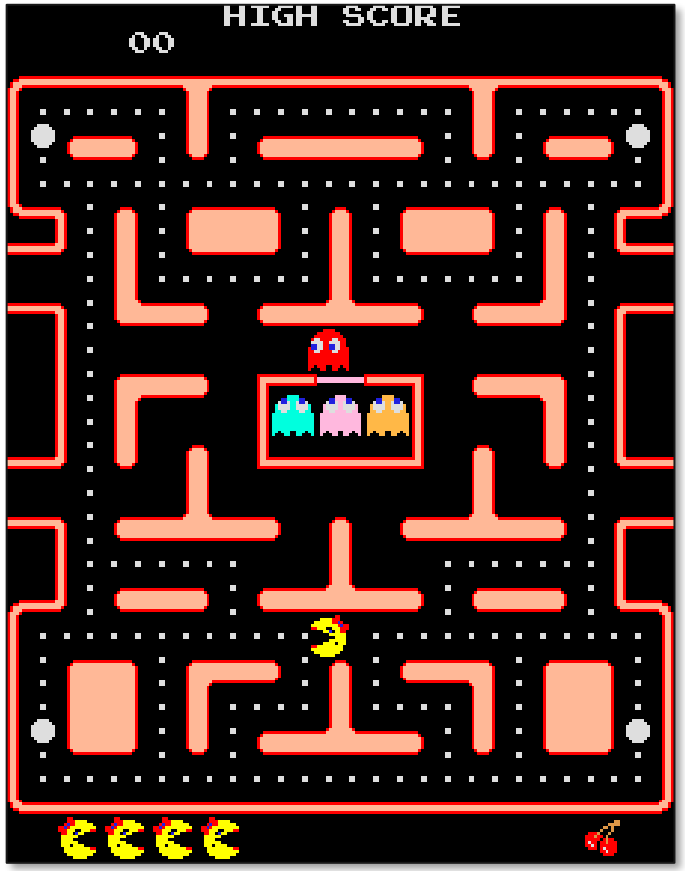


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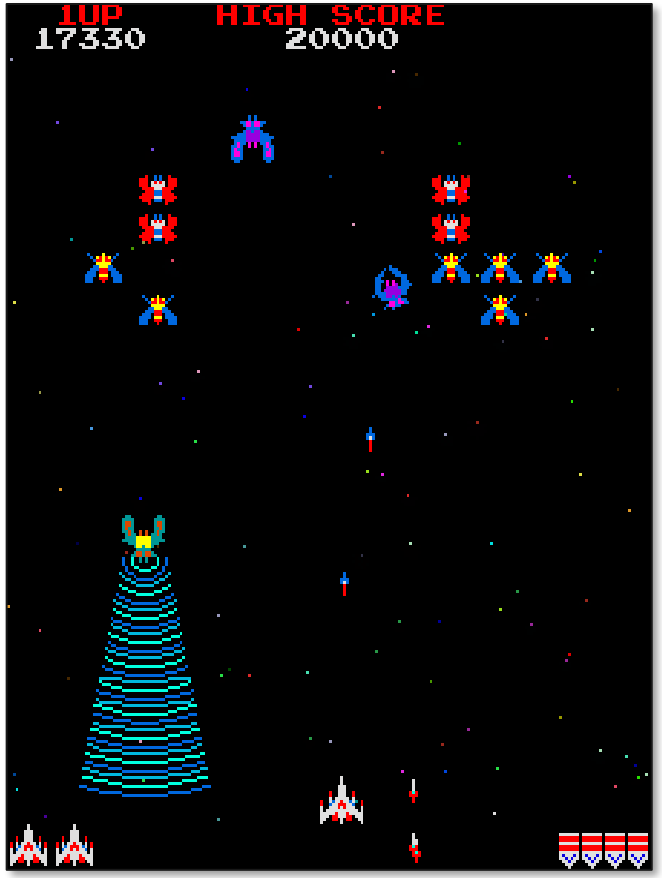


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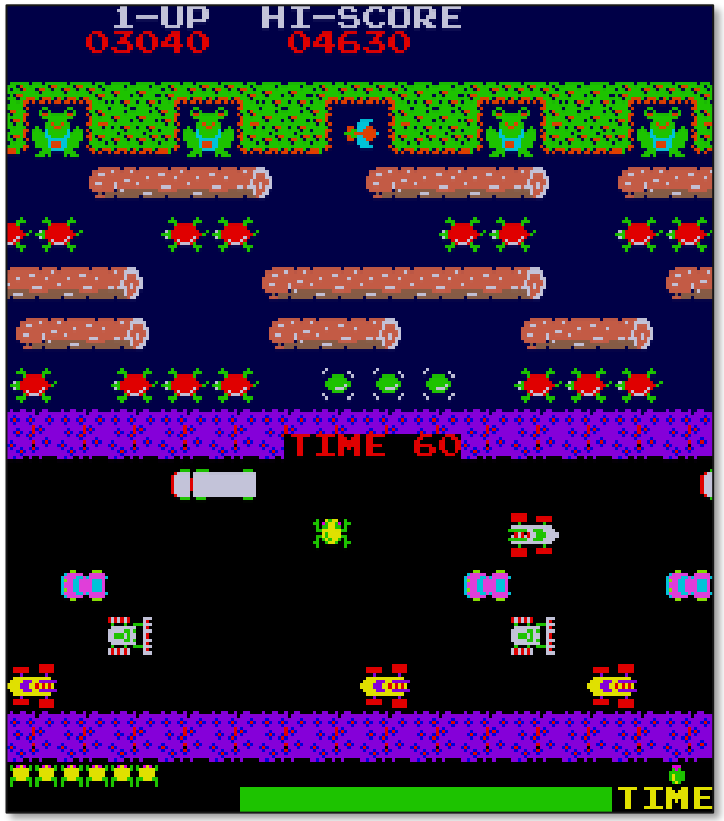


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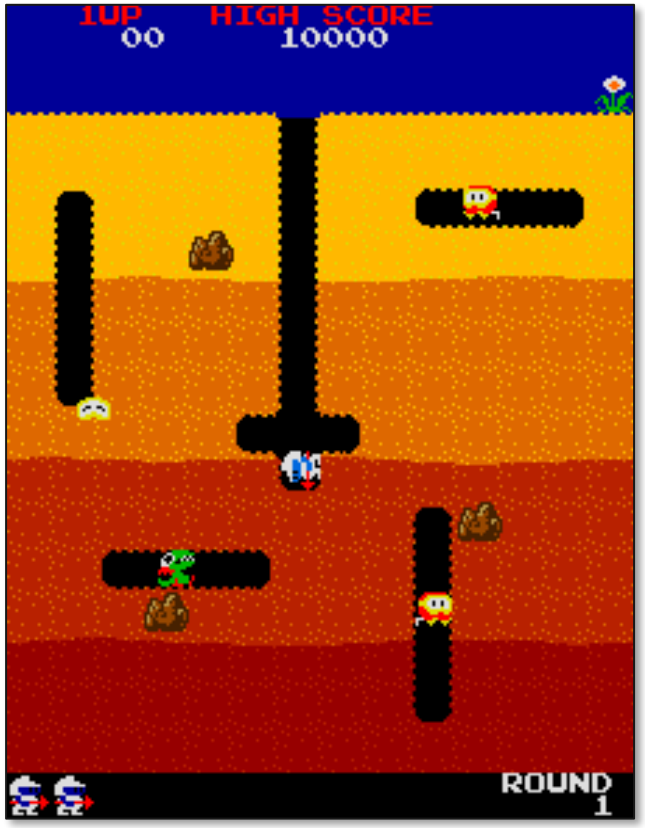


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TRS-80



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Zenith Z-89 / Heathkit H89



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Timex / Sinclair ZX81



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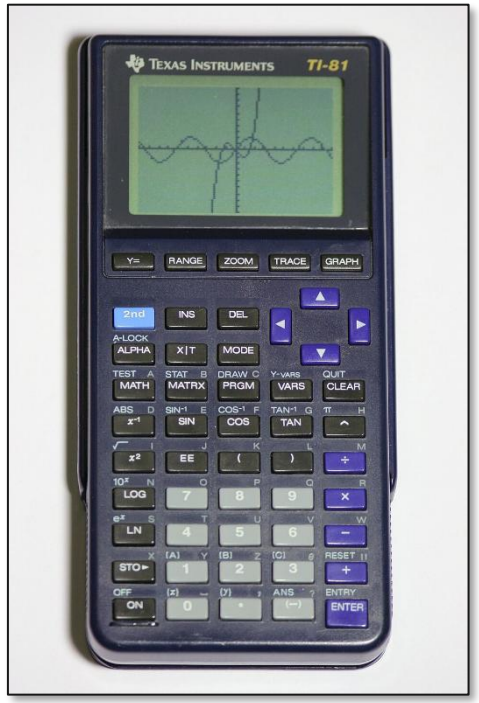
DEC Rainbow 100



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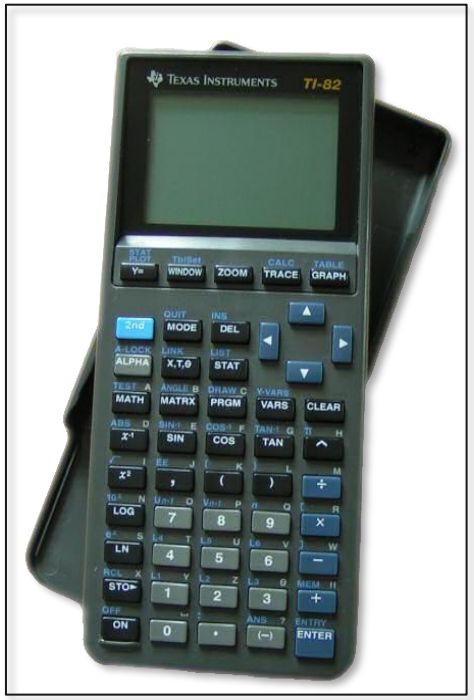
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TI-81



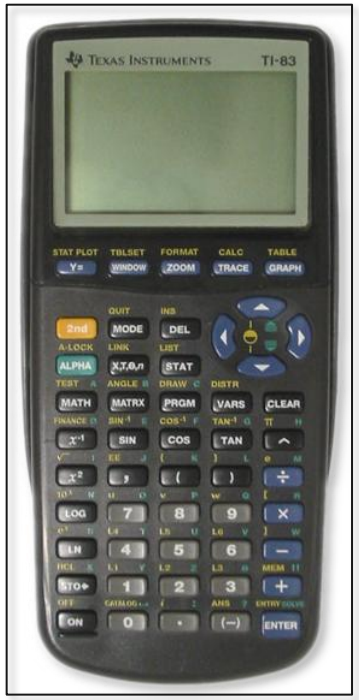
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TI-82



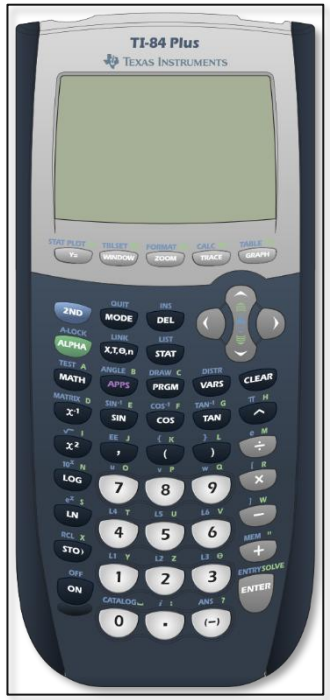
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Zilog Z80 Common Peripherals

PIO Parallel I/O Controller

CTC Counter/Timer Circuit

SIO Serial I/O Controller

DMA Direct Memory Access Controller

DART Dual Asynchronous Receiver / Transmitter



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